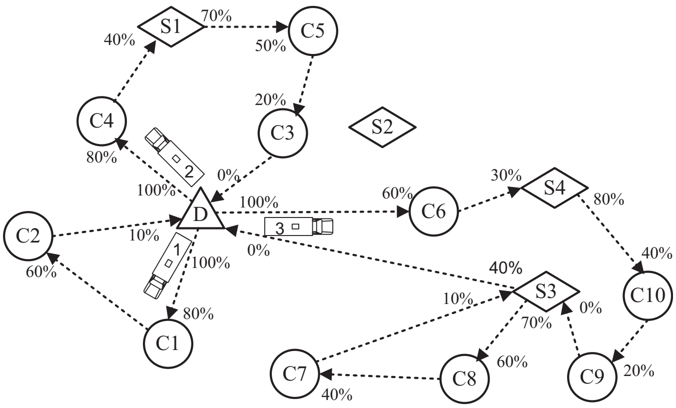

Fig. 1. An illustrative example.

with a distance d_{ij} and travel time t_{ij}. The battery charge is consumed at a rate of h and every traveled arc consumes $h \cdot d_{ij}$ of the remaining battery. Each customer $i \in V'$ has positive demand q_i, service time s_i and time window $[e_i, l_i]$. All EVs have a load capacity of C and a battery capacity of Q. At a recharging station, the battery is charged at a recharging rate of g. The decision variables, τ_i, u_i and y_i keep track of the service starting time, remaining cargo level and battery state of charge on arriving to station $i \in V'_{0,N+1}$, respectively. The binary decision variable x_{ij} takes value 1 if arc (i,j) is traversed and 0 otherwise. In Schneider et al. (2014) the battery is always recharged to full capacity, i.e. the recharge amount is $(Q - y_i)$. EVRPTW-PR allows partial recharges by defining a new decision variable Y_i which represents the battery state of charge on departure from station i.

$$\min \sum_{i \in V'_0, j \in V'_{n+1}, i \neq j} d_{ij} x_{ij} \tag{1}$$

subject to

$$\sum_{j \in V'_{n+1}, i \neq j} x_{ij} = 1 \quad \forall i \in V \tag{2}$$

$$\sum_{j \in V'_{n+1}, i \neq j} x_{ij} \leqslant 1 \quad \forall i \in F' \tag{3}$$

$$\sum_{i \in V'_0, i \neq j} x_{ij} - \sum_{i \in V'_{n+1}, i \neq j} x_{ji} = 0 \quad \forall j \in V' \tag{4}$$

$$\tau_i + (t_{ij} + s_i) x_{ij} - l_0 (1 - x_{ij}) \leqslant \tau_j \quad \forall i \in V_0, \forall j \in V'_{n+1}, \quad i \neq j \tag{5}$$

$$\tau_i + t_{ij} x_{ij} + g(Y_i - y_i) - (l_0 + gQ)(1 - x_{ij}) \leqslant \tau_j \quad \forall i \in F', \forall j \in V'_{n+1}, \quad i \neq j \tag{6}$$

$$e_j \leqslant \tau_j \leqslant l_j \quad \forall j \in V'_{0,n+1} \tag{7}$$

$$0 \leqslant u_j \leqslant u_i - q_i x_{ij} + C(1 - x_{ij}) \quad \forall i \in V'_0, \forall j \in V'_{n+1}, \quad i \neq j \tag{8}$$

$$0 \leqslant u_0 \leqslant C \tag{9}$$

$$0 \leqslant y_j \leqslant y_i - (h \cdot d_{ij}) x_{ij} + Q(1 - x_{ij}) \quad \forall i \in V, \forall j \in V'_{n+1}, \quad i \neq j \tag{10}$$

$$0 \leqslant y_j \leqslant Y_i - (h \cdot d_{ij}) x_{ij} + Q(1 - x_{ij}) \quad \forall i \in F'_0, \forall j \in V'_{n+1}, i \neq j \tag{11}$$

$$y_i \leqslant Y_i \leqslant Q \quad \forall i \in F'_0 \tag{12}$$

$$x_{ij} \in \{0, 1\} \quad \forall i \in V'_0, \forall j \in V'_{n+1}, i \neq j \tag{13}$$

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The objective function (1) minimizes total distance traveled. Constraints (2) and (3) handle the connectivity of customers and visits to recharging stations, respectively. The flow conservation constraints (4) enforce that the number of outgoing arcs equals to the number of incoming arcs at each vertex. Constraints (5) and (6) ensure the time feasibility of arcs leaving the customers (and the depot), and the stations, respectively. Constraints (7) enforce the time windows of the customers and the depot. In addition, constraints (5)–(7) eliminate the sub-tours. Constraints (8) and (9) guarantee that demand of all customers are satisfied. Constraints (10) and (11) keep track of the battery state of charge and make sure that it is never negative. Constraints (12) determine the battery state of charge after the recharge at a station and make sure that the battery state of charge does not exceed its capacity. Finally, constraints (13) define the binary decision variables.

Proposition 1. *If an optimal solution exists such that an EV leaves the depot with its battery partially charged, i.e. $Y_0^* < 1$, then the same EV departing from the depot fully charged is also optimal, i.e. $Y_0^* = 1$ is also optimal.*

Proof. Let $Y_0^* < 1$ be optimal. Since fully recharging the battery at the depot does not delay the departure time of the EV, $Y_0^* = 1$ must also be optimal. □

Corollary 1. *If $Y_0^* < 1$ is optimal, then the problem has infinite multiple optima.*

Proof. Let $\overline{Y}_0 < 1$ be optimal and $\overline{Y}_0 + \varepsilon \leqslant 1$ not, where ε is a small positive scalar. Then following Proposition 1, multiple optima exist such that $\overline{Y}_0 \leqslant Y_0^* \leqslant 1$. □

Proposition 2. *If an optimal solution exists such that an EV has been recharged at least once and returns to the depot at the end of its route with positive battery state, i.e. $y_{n+1}^* > 0$, then its return to the depot with empty battery is also optimal, i.e. $y_{n+1}^* = 0$ is also optimal.*

Proof. Let $y_{n+1}^* > 0$ be optimal. Since recharging the battery with less energy at the preceding station does not delay the departure time to cause any time window infeasibility, $y_{n+1}^* = 0$ must also be optimal. □

Corollary 2. *If $y_{n+1}^* > 0$ is optimal, then the problem has infinite multiple optima.*

Proof. Let $\bar{y}_{n+1} > 0$ be optimal and $\bar{y}_{n+1} + \varepsilon$ not, where ε is a small positive scalar. Then following Proposition 2, multiple optima exist such that $0 \leqslant y_{n+1}^* \leqslant \bar{y}_{n+1}$. □

Without loss of generality, we assume that an EV departs from the depot with a battery charged in full and returns to the depot with its battery fully consumed if it has been recharged at least once along its route.

4. Solution methodology

We propose an ALNS method to solve the EVRPTW-PR. ALNS was introduced by Ropke and Pisinger (2006a) as an extension of the Large Neighborhood Search (LNS) framework put forward by Shaw (1998). Since local search methods can only make small changes to an existing solution their search space is narrow. Hence, they are unable to move from one promising area to another within the feasible region. To overcome this shortcoming, Ropke and Pisinger (2006a) considered large moves that rearrange up to 40% of the vertices instead of using small moves that relocate or exchange only a few arcs or vertices at each iteration. In a subsequent study, Ropke and Pisinger (2006b) developed a unified ALNS heuristic for a large class of VRP with Backhauls. Pisinger and Ropke (2007) improved this heuristic with additional algorithms and showed that the proposed framework gives competitive results in different VRP variants. Since then, ALNS has been successfully implemented to solve various VRPs, e.g. cumulative capacitated VRP (Ribeiro and Laporte, 2012), pollution-routing problem (Demir et al., 2012), two-echelon VRP (Hemmelmayr et al., 2012), pickup and delivery problems with transshipment (Qu and Bard, 2012) and with vehicle transfers (Masson et al., 2013), VRP with multiple routes (Azi et al., 2014), periodic inventory routing problem (Aksen et al., 2014), and production routing problem (Adulyasak et al., 2014).

4.1. Overview of the proposed ALNS approach

4.1.1. Initial solution construction

The initial solution is obtained by iteratively constructing feasible routes. The route construction begins with the nearest customer to the depot. Then, the insertion costs of all unassigned customers to all possible existing positions in the route are determined respecting the time window constraints, i.e. the insertion of customer i between nodes j and k is calculated as